Virtual Reality and Augmented Reality and Their Effect in improving Rehabilitation Rates of Patients Suffering from Muscle Atrophy after Long-Term Bed Rest

Kenny Peng

George Mason University

Introduction

In current times there has been a rise in popularity with development of virtual reality (VR) and augmented reality (AR) systems, which is a revitalization of the original boom in the nineties. Newly introduced products such as head-mounted devices (HMD) like the Oculus Rift and the Microsoft HoloLens are reducing costs and bringing attention to the technology. While currently mainly used for video games, this new technology offers many capabilities which can be utilized in current day jobs with its screens able to display almost anything a computer can, which allows for active visualization of virtual data and objects in the real world with AR and full immersion with VR. An example of this in application for the medical field is using AR to overlay images of autopsy scans on top of viewed physical specimens allowing for faster identification of biopsy clips, which locate areas of interest in autopsy specimens. (Hanna et al., 2018) VR has also been found to be useful in helping spinal cord injury patients relearn how to drive by offering an immersive experience only capable of being provided by VR. (Carlozzi et al., 2013) Research is currently scattered among the medical field into many sub-fields ranging from education, surgery, and specifically to this proposal, rehabilitation (Dixon et al., 2012). As it is a growing field, multiple areas are relatively unexplored, and rehabilitation has been one of these areas that seems worthy of further exploration.

According to recent research, VR has been found to be effective in reducing pain of a patient when used during treatment. The reason suggested for this is that the amount of visual information provided through simulations can overload the brain making patients less able to process the neural signals which cause pain (Hoffman et al., 2014). Research has also found that AR, when used in treatments deigned for rehabilitation of stroke patients, makes patients feel motivated (Alamri et al., 2010). Researchers are constantly trying it find new benefits to AR and

VR use in their research, yet they are often focusing too much on one aspect and not capitalizing on the other prior found benefits as much as they could. An example of this is the lack of consideration of the cost and continence benefit of VR when research compared a traditional three screen simulation method to VR. (Carlozzi et al., 2013) These benefits of VR are ignored and thus makes both their research and their comparison of driving rehabilitation methods incomplete. The multitude of observed benefits of AR and VR have many uses in addition to the intended ones the studies were originally carried out for, and thus, I am seeking to combine multiple aspects of discovered benefits of VR and AR such as the motivational aspect as well as the portability aspect and apply it to a treatment to improve the rate of rehabilitating patients suffering from muscle atrophy.

In this proposed research, the question examined will be VR and AR and their effect in improving rehabilitation rates of patients suffering from muscle atrophy after long-term bed rest through a proposed trial and by looking into the ethical concerns of AR and VR that stem from its use. Both effectiveness of VR and AR use and consideration of the ethics concerning use are needed to determine future viability of rehabilitation treatment with VR and AR. Prior studies are sparse, and the relevance of the research that exists on this topic is limited by age as the research becomes less relevant as technology advances. However, with the new wave of VR and AR, there has been a noticeable uptick in research done with VR and AR, however, there is still a lack of research on use of VR and AR on rehabilitative treatment. Therefore, I find it necessary to do a study on the topic as the effects may ease the lives of many in the future. The current state of VR and AR research in the medical field is a scattershot with every study seemingly claiming to be the first to do something (Hoffman et al., 2014). This goldrush for new uses seems to distract from the application of already discovered benefits, which is detrimental to the

development of useful tools based on prior research. This can be seen by the lack of research depth on any one topic found, and the seemingly endless supply of research on other uses in the medical field. Thus, multiple studies' examinations of peer works in the specific use case are often quite short or even non-existent. (Hanna et al., 2018). It is important for research to be done at more depth to allow for productive research and conversations for continuing the development of VR and AR for use in rehabilitation.

Literature review

Current Literature for the medical applications of VR and AR and rehabilitation is mainly found within peer reviewed medical journals such as *Surgical Endoscopy, Archives of Pathology & Laboratory Medicine, and Journal of Physical Therapy Science.* The scholars conducting the research are MDs, medical researchers, medical professors, and medical fellows from hospitals, specialized research laboratories, and university departments. Their reasons for conducting this research are mainly to test the use of VR and AR in their respective fields.

What Are the Found Benefits and Viable Uses Found So Far?

For the most part, researchers are most interested in the immediate benefits or dangers of AR and VR to their respective fields which tend to vary greatly. However, some do consider the broader possibility of use within the medical field, but these tend to come with a warning as to the generalized use of the technology for all fields, and recommends careful analysis to whether it is safe to use (Dixon et al., 2013). This wary advice to other fields in research leads to less research on applications of the technology with prior found benefits as researchers are more focused on finding and confirming what benefits materialize in their own field as prior research

is hesitant to confirm usefulness in fields other than their own. Though, none of the prior research denies that AR and VR are a powerful and transformative tool if used correctly.

One study attempted to use AR for rehabilitating stroke patients and detailed the creation of the software activities intended for the use of rehabilitation, and then conducted a study to determine if the use of AR would be useful. The study tested two of the five activities created for rehab of stroke patients, a shelf exercise and a cup exercise, in multiple repeated time-based trials. Data was collected in the form of both averaged time to complete tasks in groups of five consecutive trials and a questionnaire at the end asking things like enjoyment, motivation, and whether it was easy to understand with the answer choices for each question being agree, disagree, or not sure. The results from the study were all extremely positive with improvement in task completion time across the board and 12/15 subjects saying it was easy to understand, enjoyable, and motivating. A unique factor brought up in the study was the potential for the portability of rehabilitation treatments using AR, and results were unanimous with 15/15 subjects agreeing on portability (Alamri et al., 2010).

Another study indicated that AR was useful in rehabilitation as well, but in this case the group studied were elderly women over the age of seventy. They used a study method similar to the Alamri in that they conducted a physical trial experiment, however their methods and data collected differed on content as it was a more comprehensive study that also compared the use of AR with traditional methods. Data collected was in the form of Berg Balance Scale scores, which measures balance, along with factors such as walking gait length and speed. The Otago leg exercise method, which consists of 40 minute sessions 3 times a week where exercises like walking backwards and walking heel to toe are done, was the main method used as the treatment for patients both with and without AR, and the results were positive for measurements of the

tested variables such as gait and balance for both traditional Otago and AR assisted Otago. However, the results with the use of AR were consistently better than Otago treatment without AR assistance. This is particularly interesting since it implies that age is independent of improvement as Alamri's younger, college aged study group also saw improvement. Hence AR is a useful technology for treatment of weakened muscles for all age groups (Yoo et al., 2013).

In alternate fields than rehabilitation, the use of AR was also tested with positive results in pathology. A study testing the use of AR integrated into the regular routine of autopsy in a lab, such as by using it for identifying markers for coregistration of radiographs to the physical specimens, found that the technology has advanced far enough to be comfortable and helpful for the tasks such as imaging the autopsy and visualizing the information needed to perform the autopsy such as instructions and annotations on radiographs in real time. However, their analysis of current software for use found that it was lacking for some necessary uses. The results from the study were in agreement with the other studies done in that the AR increased speed of tasks performed. Their particular mention of the factor of comfort and lack of simulation sickness, the nausea that occurs with prolonged use of simulations, is suggestive of advancements preventing the nausea from simulation sickness and prompts further research as the particular tool they're using the Microsoft HoloLens is far more recent than what other studies have used (Hanna et al., 2018).

The use of VR is also a useful alternative treatment for altering brain responses. Another study did a case study testing VR to overload the brain with information as to distract a patient from pain response signals during burn treatment. The data from the patient's response on a survey after each treatment found that they felt completely immersed in the simulation with a presence rating of 10/10 with 10 be completely immersed and 0 being not at all immersed, and

they had a consistent pain reduction when using VR during treatment. This brings up another unique use of the technology for treating patient problems involving barriers related to the brain. The possible potential is that the use of VR and AR may be able to provide motivation and mental relief to tasks deemed repetitive, but necessary to treatment and rehabilitation (Hoffman et al., 2014).

What concerns have been slowing adoption?

A recent study had a marked concern about the adoption of AR and VR for highly attentional roles such as in surgery where the fear was that there would be inattentional blindness, blindness to issues normally detected caused by lack of attention, from reliance on AR. A experimental study was done with trials involving experienced surgeons either using AR or traditional methods to perform a practice operation. The inattentional blindness was tested based on the identification of complications in the task. Results from the study showed abysmal detection rates with 0% detecting a critical complication compared to 41% in the control group with similar results for the rest of the trials. This brings to attention the issues of adopting new technology, and warn of the fact that not only are the benefits not discovered yet, but also the dangers of the AR and VR. Other literature rarely talks about risks when dealing with AR and VR, and it is suggested that this must change, and a proper analysis of benefits versus risks must be done before application of AR and VR (Dixon et al., 2013).

Previously, an older study comparing three-screen and VR treatment for rehabilitating patients suffering from spinal cord injuries to drive found that simulation sickness, was a large factor in their hesitation to recommend VR for rehabilitation with 19.2% of participants in the study falling ill from it and only 7.7% felt acute illness from the three-screen display. In

conjunction with Hanna's study this seems to be marked improvement of usability of the technology for patients. However, even with simulation sickness as a factor, the researches determined that the realism benefits of the VR were worth it (Carlozzi et al., 2013).

Why Muscle Atrophy is a Problem to Be Focused on?

A study by Parry and Puthucheary brings attention to the concerns of the inherent risks of long term stays in intensive care units (ICU). In their study they analyze prior research on the different medical issues that come from long term stays in the ICU which culminates into "post-intensive care syndrome", which is marked by muscle weakness, impaired physical functioning as well as neurocognitive and psychiatric symptoms. Their research finds that within the first week of immobilization, patients can lose up to 40% of their muscle strength to muscle atrophy and 1% of their bone density. The issue they find with current ICU use, is that it focuses on sedation of patients, which lead to leads to a decrease in muscle function from both the sedative drugs and from muscle atrophy. Specifically, for skeletal muscle mass, they find that for a 70kg person 280g of protein are lost a day and 30% of skeletal muscle mass is lost within after the first 10 days. A problem found with current rehabilitation of these patients is that outside of dedicated rehabilitation time, at least a third of patients become inactive. The researchers of this paper advise for implementing new ICU rehabilitation methods that focus on active treatment (2015).

Another study comes up with similar conclusions as with Parry and Puthucheary and also examines prior research in evaluating the harm of long-term ICU stays. However, in their case, they focus in on the musculoskeletal system and cardiovascular system deterioration. From their research, 1% to 1.5% of musculoskeletal strength is lost per day after strict bed rest, and it is even worse with cast immobilization with a loss of 1.3%-5.5% of musculoskeletal strength per

day. They also find that after 3 to 4 weeks of bed rest, resting heart rate was reported to increase by one half beat per minute and a 30% decrease in stroke volume for blood. This is attributed to the atrophy of the heart muscle similar to the musculoskeletal system due to inactivity. They, like Parry and Puthucheary, also prescribe a treatment by reforming how ICU treatment is done. However, they offer a novel idea of using prerehabilitation, where a patient before entering ICU is trained so that they have less of an impact to muscle functionality after ICU treatment by training muscles beforehand, to lower impact of ICU stays.

A different article takes a different approach from the previous ones mentioned here and analyzes the benefits of bed rest treatments in addition to the complications that occur from it before finally, as with each previous study on muscle atrophy involving the ICU in the literature review, suggesting ways to lessen the adverse effects. The researcher finds that after certain surgeries like liver biopsy, short-term bed rest can help reduce the bleeding complications. Bed rest was also found to helpful in reducing fetal complications within mother's that suffered from preeclampsia. However, other situations where bed rest is recommended usually such as in Pulmonary Tuberculosis, no benefit was found. Issues they found with bed rest treatment included things like muscle weakness, which were reported years after leaving bed rest, systemic inflammation caused by the loss of IL-6 cytokines which are produced during exercise. To partially prevent these issues, it is suggested to add resistance exercises to the treatment of bed rest patients and passive stretching, which is flexing normally restrained muscles such as those kept in a cast, daily. (Brower., 2009).

Another study tracked the recovery of acute respiratory distress syndrome throughout the year in a 3,6, and 12 month intervals to find the long term outcome of the condition. They claim that survivors of the syndrome are at risk from physical and neuropsychological complications

and seek to assess the physiological, functional, and quality-of-life of patients after ICU discharge. They gathered 109 subjects with varying levels of the syndrome and at each interval they were interviewed and given a walking test. The results they discovered from their experiment was that the patients had improved in health, but slower than expected. Patients also had a 11% mortality rate with more of it within the first 6 months. An explanation for the loss of functional ability and quality of life is impaired muscle function. They conclude that a year is not enough to recover fully, and that they are unsure of how long or if it can even be cured. At the end of their research they concluded similarly to all the previous research in this literature review relating to muscle atrophy from ICU stays, that there is a need for more research on muscle atrophy from ICU stays and research on new practices to prevent muscle atrophy both in ICU and after discharge (Herridge et al., 2003).

Analyzing Potential for Application Today

Overall research has had varying goals, but there has been a lack of true application of findings from research, and instead a focus on finding the benefits of VR and AR. Current research prompts the application of the previously found benefits as well as a minding of the dangers found. Furthermore, some of the research has been limited by old technology, and their concerns such as simulation sickness must be analyzed with current technology to examine if they still exist. Therefore, the question proposed will be investigating the effectiveness of AR and VR with the benefits that research found beforehand and analyzing the current state of previously found issues using current technology.

This study will be contributing to the integration of VR and AR in the rehabilitation of bedridden patients and improving their recovery speed and general wellness of their lives. In

addition, this study will provide beneficial insight on the current state of the technology and its usability in other fields of medicine as well not limited to rehabilitation. Previous bias from small or select samples used for subjects during research will also be checked with this study.

Methods

The methodology of the research that will be followed will be similar to that of previous research with the difference being the content of the research as well as the tools used for VR and AR. As with previous research an experimental method with a group of subjects will be the choice for the experiment as that is the only practical way to collect data involving rehabilitation rates. The subjects in the experiment would consist of 45 male post-ICU patients between split into 3 groups of 15. The 3 groups will be based on time spent in ICU with group 1 spending one week in the ICU, group 2 spending two weeks in the ICU, and group 3 spending three weeks in the ICU. Furthermore, these three groups will be split into 3 further subgroups of 5 with one doing rehabilitation activities with AR, one with VR, and one control group doing the activities without VR or AR as the control group. These three groups and subgroups are needed to compare the benefits of VR and AR to traditional methods of rehabilitation for data and the effect of different levels of muscle atrophy to see if recovery rates differ based on level of atrophy. The number of subjects is based on prior research which share a similar number of subjects (Carlozzi et al., 2013). and on the fact that it would be difficult to collect a larger number of ICU patients with the time conditions needed. The tools that will be used are the Oculus Rift VR HMD, which is chosen due to its superior head tracking capabilities compared to competitors, which is important in reducing simulation sickness and the Microsoft HoloLens AR HMD, chosen for its computing power capabilities needed to run the activities for the trials (Hanna et al., 2018). Software would also need to be created to have similar activities to

traditional methods of rehabilitation as the software like that doesn't exist yet, and it will be similar to the AR-REHAB software that Alamri designed for his experiment involving activities like picking up virtual cups (2010).

The methods for recovery in this experiment will be a set of rehabilitation activities similar to the generic routine suggested by Topp for rehabilitation, which involves slow walking for 3–5 minutes increasing by 1 minute when tolerable to 10 minutes to strengthen leg muscles and warm up before other rehabilitation activities. Aerobic exercises will be done such as going up stairs or walking depending on capability and having heartrate going up by 10-15 beats per minute over resting for 5 minutes 2-3 times a week and eventually increasing intensity and duration as tolerated to 20-25 beats per minute for 30 minutes, 5 days a week. The biceps curls, overhead press, and Triceps extensions with 6-8 repetitions twice per week increasing as tolerated to 5 days per week with increasing weights (2012). The rehabilitation plan that is used for this study is a simplified version focusing on leg and arm muscles to simplify data collection as other muscle areas are more difficult to accurately measure. When the experiment begins, the VR and AR groups would receive their respective HMD's and the control would do the rehabilitation without AR or VR. The AR and VR groups will be using their HMD with software to "gamify", or make their rehabilitation activities seem more like a game similar to how Alamri used AR and Hoffman used VR (2010 & 2014).

The methods for collecting data will be through both physical tests and surveys. Data will be collected in intervals with 0, 3, 6, and 12 month periods based off of Herridge's. The physical tests will also be akin to the walking test done by Herridge, but instead, this study will be doing a mile run as to be able to examine a more full recovery (2003). This is because using the ability to walk to determine full recovery has its limitations as there is a point where you can no longer see

improvement in walking ability. To test the upper body for muscular strength and endurance, pushup tests will also be conducted to test arm muscles. Pushups were chosen as they can be used to track both endurance and strength. Muscle mass will also be measured by measuring lean body mass, body mass with fat subtracted, at each data collection interval. This would be to measure the growth of muscles. A survey will also be given out at each interval similar to Alamri and Hoffman's, which asks for motivation level, comfort, level of simulation sickness experienced, and convenience from a scale of 1-10 (2010 & 2014). Motivation, comfort, convenience, and level of simulation sickness experienced are asked as they are important factors in willingness to continue rehabilitation treatment and the effort in participating in treatment.

Data collected will be analyzed for rate of improvement between AR, VR, and control group to provide an overview of the effects of VR and AR when used to augment rehabilitation compared to traditional rehabilitation. This will be done by statistical analysis and scatter plot graphing the results of the physical tests with the results from the sub groups of AR, VR, and control separately averaged with a line of best fit to compare difference of recovery rate based on the use of AR, VR, and nothing. This will reveal if AR and VR are beneficial in increasing the recovery rate of muscle atrophy patients. Furthermore, this would be done for each of the main groups that were separated by time in ICU. Then the averaged subgroups from each of the main groups would be compared to each other. For example, the 3 graphs for VR from each main group will be compared against each other to observe trends of recovery rates based on time in ICU. The survey data will be compiled into bar graphs, which would again, be separated by subgroup among weeks spent in ICU to see if they maintain consistency among results. Simulation sickness will be analyzed based on average level of it within the AR and VR subgroups among the intervals for considerations on the technologies viability for use. The

opinion of VR and AR treatments over time is an important consideration collected as the patients may be biased early on based on the new experience of using AR and VR. If the results are consistent among the intervals however, the opinions of use can then be deemed accurate.

Conclusion

This research will be contributing to the integration of VR and AR in the rehabilitation of bedridden patients and improving their recovery speed and general wellness of their lives. As could be seen from the trends of the conclusion of research involving muscle atrophy, the need for a transformation of the methods of taking care of muscle atrophy is desperately needed This technology has the potential to improve and change how all rehabilitation is done, not only post-ICU patients but other types of rehabilitation as well such as for stroke and spinal cord injuries. VR and AR also have the potential to make the process more affordable and available to those that need it as well since the devices are portable and cheaper than professional treatment. An issue faced by many is the lack of access to rehabilitation treatments whether its because of distance to urban centers or due to the cost and VR and AR may be able to remedy this. The application of this research, if successful, may also lead to more comprehensive recoveries as the current standard for declaring someone rehabilitated is a rather low bar as can be seen by studies cited in this proposal with patients complaining about muscle weakness long after discharge. Future research would hopefully expand this research onto other types of rehabilitation practices and examine the long term effects and results of VR and AR in rehabilitation as that can only be comprehensively done after a technology has gained wide spread acceptance. Upcoming research would also be well advised to continue examining developments in VR and AR as the technology is rapidly improving and may see major changes and improvements in the near future.

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